

Acoustic Correlates of the Spanish Tap-Trill Contrast: Heritage and L2 Spanish Speakers

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ABSTRACT

The present study investigates the acoustic correlates of the Spanish tap-trill phonological contrast (/ɾ/-r/) in the production of 40 Spanish heritage speakers and 20 L2 Spanish learners in Northern California. The acoustic analyses examined the number of occlusions and overall duration in the production of phonemic trills, while the phonetic variants of the phonemic tap were based on the degree of apical constriction: true tap, approximant tap, and perceptual tap. The results from a reading-aloud task indicate that most speakers produced non-canonical phonemic trills with one or zero occlusions and maintain the Spanish tap-trill phonological contrast largely by means of segmental duration, and that this is especially true for L2 learners and English-dominant heritage speakers. In contrast, Spanish-dominant heritage speakers produced the majority of their trills with two or three brief occlusions between the tongue apex and the alveolar ridge. These data confirm that heritage speakers are a heterogeneous group and that variance in their rhotic production is a result of language dominance: English-dominant heritage speakers and L2 learners are most likely to exhibit a modified system to maintain the rhotic phonological contrast in comparison to Spanish-dominant heritage speakers. The findings of this study add to our understanding of the sources of variation in heritage and L2 pronunciation by investigating a largely understudied bilingual population that has traditionally been ignored in bilingual phonetic research.

Keywords: *rhotics, language dominance, heritage speakers, L2 speakers, tap-trill contrast*

INTRODUCTION

Heritage speakers are early bilinguals who have been exposed to the minority (i.e., heritage) language and the majority language early in life,¹ either by growing up speaking both languages since birth (i.e., simultaneous bilinguals) or having been brought up in a monolingual setting in early childhood and becoming bilingual after starting school in the majority language at around the age of 5 or 6 (i.e., sequential bilinguals). The linguistic abilities of heritage speakers have been compared in recent years to both monolingual speakers and to second language (L2) learners, with the results of relevant studies showing that heritage language acquisition typically results in a non-native-like competence and use of the language in the form of transfer from the dominant language, a better ability with receptive than productive language, and linguistic gaps that resemble the patterns attested in L2 acquisition (Kondo-Brown, 2004; Montrul, 2011; Montrul, Bhatt, Bhatia, & Girju, 2012; O'Grady, Kwak, Lee, & Lee, 2011; Polinsky, 2006; Rothman, 2007).

Although in the last few decades there has been an increase in the number of studies investigating the linguistic abilities of Spanish heritage speakers, heritage Spanish phonetics and phonology has received considerably less attention than other linguistic subfields, especially in comparison to research focusing on heritage language morphosyntactic knowledge and studies

aimed at devising pedagogy tailored to meet heritage language learners' linguistic needs (Potowski, 2013; Rao & Ronquest, 2015). The general assumption, however, has been that Spanish heritage speakers have a benefit in pronunciation as a result of early exposure to the minority language (Au, Oh, Knightly, Jun, & Romo, 2008; Chang, Haynes, Rhodes, & Yao, 2008; Knightly, Jun, Oh, & Au, 2003), but at this point we cannot claim that we clearly understand the phenomenon of Spanish heritage language phonological acquisition since there are still very few studies that have empirically investigated the Spanish pronunciation of these heritage speakers (see Amengual, 2012; Boomersshine, 2012; Henriksen, 2015; Kim, 2015; Rao, 2014, 2015; Ronquest, 2012). Since studies have shown differences between the pronunciation of heritage speakers and monolingual speakers of the minority language (Au, Knightly, Jun, & Oh, 2002; Bullock, 2009; Chang, Haynes, Yao, & Rhodes, 2009; Ronquest, 2012) we must question the bold and unchallenged assumption that heritage speakers manifest an intact phonological system.

According to U.S. Census data, the Hispanic/Latino population in the United States has reached 53 million, not counting approximately 3.6 million residents of Puerto Rico (U.S. Census Bureau, 2015a), making it the nation's largest minority, and there are approximately 38 million Spanish speakers in the U.S. (U.S. Census, 2015b). Furthermore, the Instituto Cervantes (2015) estimates that by 2050, the U.S. will have the largest number of Spanish-speakers in the world, and many of those will be heritage speakers of Spanish. The growing presence of Spanish in the U.S. being attributed to both native speakers of Spanish who emigrate from Spanish-speaking countries, principally from Mexico, as well as Spanish heritage speakers who are born in the U.S. warrants further investigation and comparison of these bilinguals' Spanish pronunciation patterns. There is ample consensus by bilingualism researchers that the nature of heritage language pronunciation is a topic that deserves much more in-depth study (Montrul, 2012). The present study follows this line of inquiry by examining the acoustic correlates of the Spanish tap-trill contrast (/ɾ/-r/) in the production of 40 heritage speakers and 20 L2 learners of Spanish. This study not only contributes to the field of Spanish dialectology, but also adds to our understanding of the sources of variation in heritage and L2 pronunciation by investigating a largely heterogeneous bilingual population that differs in age of acquisition, language experience, and language dominance.

Phonological and Phonetic Description of Spanish Taps and Trills

The Spanish voiced alveolar tap /ɾ/ and trill /r/ contrast in intervocalic word-initial and word-medial contexts. In other phonological contexts, taps and trills appear mostly to demonstrate complementary distribution, with trills appearing after a heterosyllabic consonant or word-initially and taps appearing tautosyllabically in the second position of a complex onset or word finally when resyllabified into a following vowel-initial word. In syllable final position, there is typically free variation, although the most common realization is the tap. Table 1 provides the distribution of the Spanish rhotics as a function of the phonological contexts that favor either the tap or the trill rhotic segment. The current study investigates the production of the tap-trill contrast realized in intervocalic position in word-medial contexts, which is the only phonological context that allows minimal pairs.

Table 1.*Distribution of Spanish rhotics as a function of phonological context*

Phonological context	Realization	Example
(1) #__ (word-initial) (2) C.__ (after heterosyllabic consonant)	Only /r/	/roka/ 'rock' /enredo/ 'mess'
(3) C__ (after tautosyllabic consonant) (4) V__#V	Only /r/	/broma/ 'joke' /ser o no ser/ 'to be or not to be'
(5) V__V (intervocalic)	/r/-/r/ contrast	/karo/ 'expensive' vs. /karo/ 'car' (word-medial) /aroma/ 'aroma' vs. /aroma/ 'to Rome' (word-initial)
(6) V__ (#)C (before consonant) (7) V__## (word-final before pause)	Variable rhotic	/parte/~parte/ 'part' /bailar/~bailar/ 'to dance'

Previous articulatory and acoustic studies on the Spanish taps and trills have shown that the Spanish tap is produced with a single rapid ballistic movement initiated by muscle constrictions involving contact of the tip of the tongue against the alveolar ridge, whereas most descriptions of the trill indicate that it is typically produced with two or more brief occlusions between the tongue apex and the alveolar ridge (Hualde, 2005, p. 44). Even though canonical trills have been described as involving two or three occlusions (Solé, 2002), in many languages, including Spanish, trills can be produced with as few as one occlusion or contain more than three lingual contacts (Ladefoged & Maddieson, 1996). The articulatory gesture required to produce taps and trills is different: taps are produced with a direct movement of the tip of the tongue toward the roof of the mouth (Ladefoged & Maddieson, 1996, p. 231) and trills require precise muscular control (i.e., retraction of the postdorsum and lowering of the tongue predorsum) and a critical velocity of airflow in order to initiate the appropriate vibratory movement (Widdison, 1998). Even though many varieties of Spanish display other phonetic variants of the voiced alveolar trill, such as a voiced uvular trill or an assibilated trill (Canfield, 1981; Lipski, 1994), the voiced alveolar trill is not only the most common variant across varieties of Spanish, but it is also the sound taught to students in Spanish language courses in the U.S. (Face, 2006).

The Production of Spanish Taps and Trills by Monolingual and Bilingual Speakers

Most of the research on Spanish rhotics has either focused on dialectal variation in the production of taps and trills (Colantoni, 2006a; Díaz-Campos, 2008; Henriksen & Willis, 2010; Willis, 2006, 2007, among others) or the acquisition of the tap and/or trill by English-speaking Spanish language learners (Face, 2006; Major, 1986; Olsen, 2012; Reeder, 1998; Rose, 2010). An increasingly large body of research has examined the divergent patterns of rhotic production in several dialects of Spanish that provide evidence of cross-dialectal allophonic variation. For instance, Hammond (1999) observes that the intervocalic tap-trill phonemic contrast has been lost in many Spanish varieties. Furthermore, some dialects have been shown to maintain the contrast between the phonemic taps and trills by means of overall segmental duration even when

there is no difference in the number of lingual contacts (Willis & Bradley, 2008; Bradley & Willis, 2012). First language (L1) acquisition studies have suggested that articulatory complexity triggers variation in trill production (Widdison, 1998; McGowan, 1992). For instance, instead of the canonical alveolar trill with two or more occlusions, non-trill variants that include fricatives and approximants in Argentinian Spanish (Colantoni, 2006a, 2006b) and assibilated rhotics in Highland Ecuadorian Spanish (Bradley, 2004) have been attested. Furthermore, the most common phonemic trill productions reported in Dominican Spanish are a pre-breathy-voiced tap, a pre-breathy-voiced trill, and a variant with breathy-voiced frication (Willis, 2007), and in Veracruz Mexican Spanish, reduction in the degree of lingual trilling has been shown to be followed by an approximant (Bradley & Willis, 2012). Henriksen and Willis (2010) investigate trill production in Andalusia (Spain), which shows that most phonemic trill productions in this dialect contain zero or one apical closures, including fricatives, approximants, and r-colored variants. The authors argue that individual factors may carry more weight in rhotic variation than social factors such as gender or age. In a recent study, Henriksen (2014) reports variation in the production of trills of monolingual Spanish speakers in the regions of León and Ciudad Real (Spain). Acoustic analyses indicate that these speakers produce a considerable number of non-canonical variants of phonemic trills containing less than two apical closures, including fricatives, approximants, and r-colored variants. These acoustic studies are in line with previous findings showing that many monolingual Spanish speakers maintain the phonemic distinction via segmental duration of trill variants rather than the number of lingual contacts with the alveolar region.

The pronunciation of rhotics has also received attention in the field of L2 acquisition. Major (1986) investigates the rhotic production of four beginner students of Spanish, tracking their improvement in an intensive eight-week Spanish language course. The data were obtained from the reading of a word list and a sentence list containing the target sounds in seven recording sessions spread out across the eight-week course. The results show that three out of four students improve their productions of the intervocalic tap. For the intervocalic trill, only two of the four subjects show improvement. Importantly, two of these participants not only fail to improve, but were unable to articulate the target trill at all. This study also demonstrates significant variation in the articulation of rhotics by these L2 learners: 21 different sounds were categorized for the target trill and four for the target tap. Focusing exclusively on Spanish trills, Reeder (1998) examines the production patterns of 40 L2 Spanish learners divided into four groups as a function of language proficiency: beginning (first semester), intermediate (third semester), advanced (upper-level undergraduates and graduate students), and near-native speakers (Spanish faculty). As in Major (1986), trills were considered to be accurate when produced with two or more occlusions. The results indicate a 7% accuracy rate in the beginner's data, 13% in the intermediate data, 37% in the productions of advanced speakers, and 87% in the productions of near-native speakers. As expected, an increase in accuracy correlates with language proficiency; however, it is worth noting how low trill accuracy is for all groups other than the near-native speakers. In addition to proficiency effects, Colantoni and Steele (2008) demonstrate that because sounds are easier to perceive and produce in word-initial and intervocalic positions, L2 learners are more likely to be successful in mastering the Spanish tap in intervocalic position than in other positions. Based on data from 19 L2 Spanish learners and 10 L1 Spanish speakers, their results indicate that while advanced learners show all properties associated with rhotics in

their speech (i.e., length, manner, and voicing), intermediate learners only exhibit length and manner in intervocalic position. Interestingly, intermediate learners of Spanish did not master any of these properties in word-final position.

Although several studies have examined accuracy rates in the Spanish tap and trill productions of Spanish language learners, fewer studies have investigated this contrast in the same phonological environment in order to determine whether L2 Spanish speakers differentiate these two phonemes in their production (Rose, 2010). Three recent studies analyze the productions of intervocalic taps and trills by L2 learners of Spanish (Face, 2006; Olsen, 2012; Rose, 2010). Face (2006) investigates the acquisition of the Spanish tap and trill by 41 L2 Spanish learners in two proficiency levels: 20 learners enrolled in a fourth semester Spanish course and 21 Spanish majors or minors enrolled in an upper-division elective course. As a control group, five L1 speakers of Spanish were also included in the study. The results from a reading task show that these learners acquire the Spanish alveolar tap with a high degree of accuracy but fail to succeed in the articulation of the alveolar trill. More recently, Rose (2010) analyzes the production of intervocalic tap and trills in the spontaneous speech of 24 learners of Spanish and six L1 Spanish speakers by means of two picture-story description tasks. The L2 Spanish learners were L1 English speakers enrolled in four different levels of Spanish. The results of this cross-sectional study indicate that L2 Spanish learners go through five stages before they are able to produce the intervocalic contrast in a native-like manner. Both Face (2006) and Rose (2010) demonstrate a level of instructional effect on target-like productions of Spanish taps and trills. Finally, Olsen (2012) investigates the effects that the L1 English rhotic (retroflex-like and bunched-like) has on the acquisition of L2 Spanish rhotics. The productions of 48 introductory level Spanish students reading texts that elicited Spanish /r/, Spanish /r/, and English /ɹ/ suggest that English rhotic articulation is a predictor of accurate trill production at the earliest stages of L2 Spanish acquisition. Specifically, those learners who employ retroflex articulations in English produce Spanish trills with a higher accuracy than learners that use bunched articulations. English rhotic articulation also correlates with tap accuracy rate when taking prior exposure to Spanish into consideration.

Tap and trill productions are inherently variable, and this variability, largely due to the articulatory complexity of these sounds, leads to considerable inter- and intra-speaker variation. Since previous research on monolingual Spanish speakers has shown considerable variation in phonemic tap and trill articulation, it is expected that this variability should also be found in the acoustic realization of early bilingual speakers, such as heritage speakers. Henriksen (2015) examines the acoustic correlates of the phonemic tap-trill contrast in the productions of eight first generation immigrant speakers who arrived to the U.S. after adolescence, and eight second generation (i.e., heritage speakers) of Mexican Spanish who were born and raised in the greater Chicago area. The results from a narrative task do not yield statistical differences between the first generation immigrant group and the U.S. born heritage Spanish group with regards to their pronunciation of intervocalic taps and trills. Even though there is much individual variation in the data, the results indicate that these bilinguals most commonly produce phonemic trills with a single apical occlusion, and that almost all speakers realize the tap-trill contrast by means of segmental duration, and not number of occlusions. These duration differences between taps and trills are also found in the production of low proficiency heritage speakers, intermediate

proficiency heritage speakers, and L1 speakers of Mexican Spanish in Houston, Texas (Goodin-Mayeda, 2016).

Motivated by the body of research summarized to this point, the present study examines the acoustic correlates of the Spanish tap-trill phonological contrast and compares their realizations as a function of the language dominance of 20 L2 learners of Spanish and 40 heritage Spanish speakers residing in a bilingual community in Northern California. In addition to providing data from a relatively large participant group (N=60), the present study analyzes a data set of just under 2,400 rhotic productions, a considerably higher number of tokens than those elicited in previous studies on the tap-trill contrast.

METHODOLOGY

Participants

Sixty participants were recruited (52 females, 8 males) to participate in the present study. The sample consisted of undergraduate students at the University of California, Santa Cruz. All participants reported normal speech and hearing and normal or corrected to normal vision, and each participant received a stipend for their participation in the study. The sample consisted of 40 Spanish heritage speakers and 20 L2 Spanish learners. The Spanish heritage speaker group consisted of early bilinguals who had been raised and educated in a bilingual environment in the United States, having extensive exposure to both Spanish and English on a daily basis. All participants had been raised speaking Spanish at home with Spanish-speaking parents and had learned English in pre-school or kindergarten (i.e., early sequential bilinguals) and to date had been educated in the U.S. The L2 Spanish learners had been raised in a monolingual English household in the U.S., and thus spoke English as their first language and learned Spanish at school. They reported not being L1 speakers of or fluent in any other language. The L2 learners were Spanish Studies majors, had completed their entire education in the U.S, and were enrolled in upper-division Spanish courses at the time of testing. In order to be enrolled in upper-division Spanish courses, students are required to achieve a certain score on a placement test and advance beyond the six-quarter language course sequence to a set of core upper level courses in Spanish. Each of these participants began studying Spanish after adolescence, either in high school or at the university level. These learners would be considered to be at the upper-intermediate level of university-level language study.

All participants completed the Bilingual Language Profile (BLP) questionnaire (Birdsong, Gertken, & Amengual, 2012). The BLP is an instrument for assessing language dominance through self-reports that produces a continuous dominance score and a general bilingual profile that takes into account multiple dimensions: age of acquisition of the L1 and L2, frequency and contexts of use, competence in different skills, and attitudes towards each language. All of these factors are organized in four modules (language history, language use, language proficiency, and language attitudes), which receive equal weighting. For more information, see Gertken, Amengual, and Birdsong (2014). The BLP was administered prior to beginning the experiments, and was provided in either Spanish or English, depending on participant preference. The classification of participants as Spanish-dominant or English-dominant was determined by their responses to the questionnaire, which generated a language particular score for each module, a global score for each language, and a global score of dominance. The point system was

converted to a scale score in which the Spanish score was subtracted from the English score. Dominance scores ranged from -72 to 137. Participants were assigned to groups based on their questionnaire scores. Thirteen heritage Spanish speakers obtained negative dominance scores, and thus were assigned to the Spanish-dominant group (i.e., Spanish-dominant heritage speakers, SDHS). A second group (i.e., English-dominant heritage speakers, EDHS) was formed with the 27 English-dominant participants with positive scores closer to zero. These EDHS are closer to being “balanced” bilinguals. The remaining 20 participants were assigned to the L2 learner group (i.e., English-dominant L2 Spanish, L2). These participants had positive BLP scores, far from zero, which suggests that they are more strongly dominant in English than the speakers in the EDHS group. Figure 1 provides the distribution of the three participant groups and Table 2 provides the language background for each speaker group.

Figure 1. Language dominance as a function of group according to the BLP

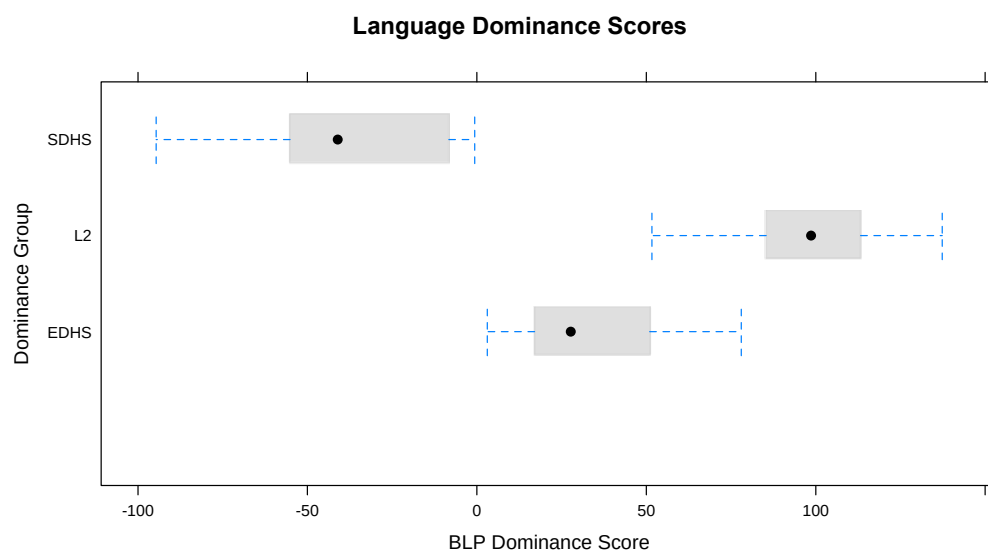


Table 2.

Age, age of exposure, accent self-ratings, and typical daily use of both languages for each group

	SDHS	EDHS	L2
	M (SD)	M (SD)	M (SD)
Age	20.3 (1.5)	20.9 (1.2)	21.2 (2)
Age of exposure	ENG=4.5 (3.6) SPN=0.2 (1.4)	ENG=1.5 (2.8) SPN=0.7 (0.6)	ENG=0 (0) SPN=6.1 (4.3)
Self-reported accent (1 = strongly accented; 9 = native-like)	ENG=8.4 (2.1) SPN=8.8(1.9)	ENG=9.1 (1.7) SPN=7.2(1.8)	ENG=9.8 (0.7) SPN=5.5(1.8)
Typical daily use (1= only English; 9= only Spanish)	7.6 (1.8)	5.3 (1.5)	3.3 (1.5)

Materials and Procedure

The production of the target voiced alveolar taps /ɾ/ and voiced alveolar trills /r/ in word-medial intervocalic position was elicited in a read-aloud task. The materials consisted of two blocks of 20 Spanish sentences. Each sentence contained a target word within the carrier phrase, *Yo puedo decir* TARGETWORD ('I can say TARGETWORD'). The Spanish utterances in each repetition block elicited the pronunciation of 10 taps and 10 trills. In addition to the rhotic data, there were 100 fillers/distractors in each block. Table 3 below provides a list of the experimental stimuli, adapted from Hualde (2014, p. 182).

Table 3.

Experimental items included in the production study

Intervocalic tap /ɾ/	Intervocalic trill /r/
<i>pero</i> 'but'	<i>perro</i> 'dog'
<i>pera</i> 'pear'	<i>perra</i> 'female dog'
<i>para</i> 'for, in order to, to'	<i>parra</i> 'vine'
<i>coro</i> 'choir'	<i>corro</i> '(I) run'
<i>quería</i> '(he/she) wanted'	<i>querría</i> '(I) would want'
<i>caro</i> 'expensive'	<i>carro</i> 'car'
<i>moro</i> 'moor'	<i>morro</i> 'snout, nose'
<i>cero</i> 'zero'	<i>cerro</i> 'hill'
<i>mira</i> '(he/she) watches'	<i>mirra</i> 'myrrh'
<i>vara</i> 'stick, cane'	<i>barra</i> 'bar, counter'

The production task was conducted individually in a sound-attenuated booth in the University of California, Santa Cruz's Bilingualism Research Lab with participants comfortably seated in front of a computer display. Participants were told that the study involved reading sentences on a computer screen and that their speech would be recorded for subsequent acoustic analysis. Each sentence was presented on a computer screen and participants were asked to read the sentences clearly and at a natural pace, speaking neither too quickly nor too slowly. The 20 sentences appeared twice and in random order. The speech samples were recorded using a head-mounted microphone (Shure SM10A) and an audio interface (MOTU Ultra Like mk3), digitized (44kHz, 16 bit quantization), and computer-edited for subsequent acoustic analysis. Each participant produced 40 target productions (i.e., 20 taps and 20 trills) for a total of 2,400 measurements (i.e., 1,200 taps and 1,200 trills). Sixteen tokens were excluded due to recording errors or mispronunciations; as a result, the dataset comprised a total of 2,384 measurements.

Acoustic and Statistical Analysis

Phonemic taps and trills were segmented in Praat (Boersma & Weenink, 2015) using synchronized waveforms and spectrographic displays. Praat scripts were run to parse each participant's recording into individual files for each target item, and text grids were created by manually marking the rhotic segment in each token. The analysis of phonemic trills consisted of measuring the number of occlusions and the overall duration of the rhotic segment (in

milliseconds, ms). The acoustic cues leading to the determination of an occlusion were the presence of lingual contacts, a reduction in the amplitude of the waveform, a transition in F3 and F4 formant structure, and changes in intensity as observed in the spectrogram (Martínez Celdrán, 1998). Figures 2 through 6 show spectrographic displays for 4-occlusion, 3-occlusion, 2-occlusion, 1-occlusion, and 0-occlusion trills, respectively.

Figure 2. Intervocalic word-medial phonemic trill production containing four apical occlusions in the word *perro* ('dog,' trill duration = 122 ms)

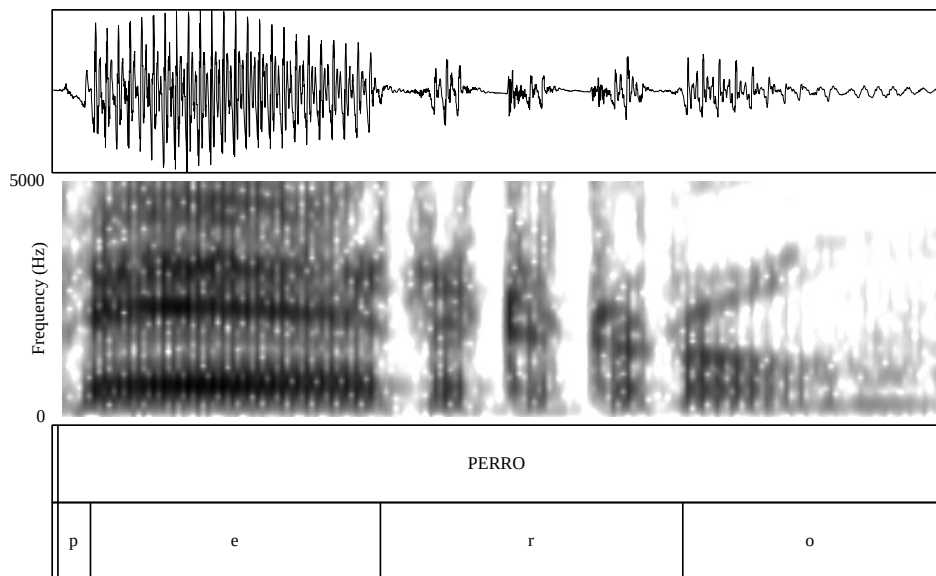


Figure 3. Intervocalic word-medial phonemic trill production containing three apical occlusions in the word *morro* ('snout, nose,' trill duration = 110 ms)

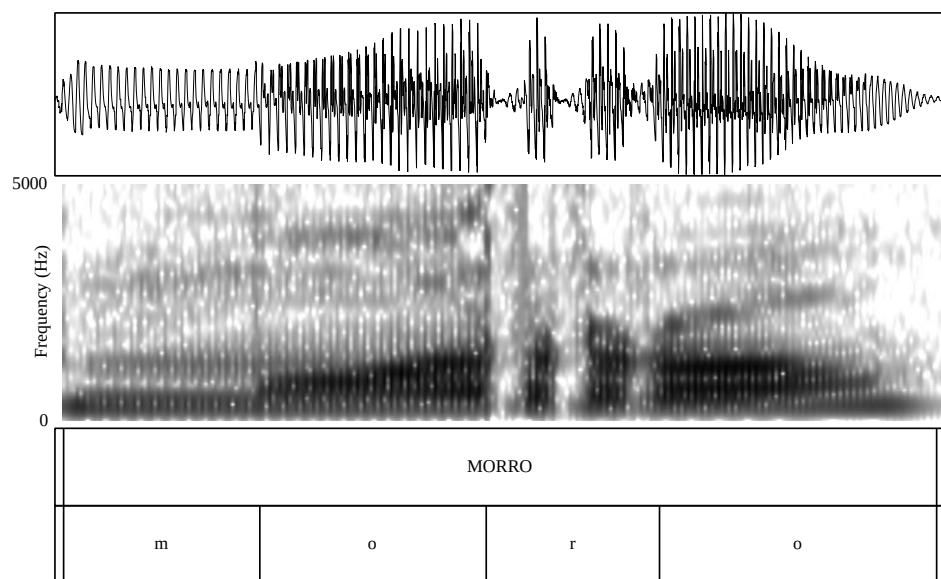


Figure 4. Intervocalic word-medial phonemic trill production containing two apical occlusions in the word *barra* ('bar, counter,' trill duration = 77 ms)

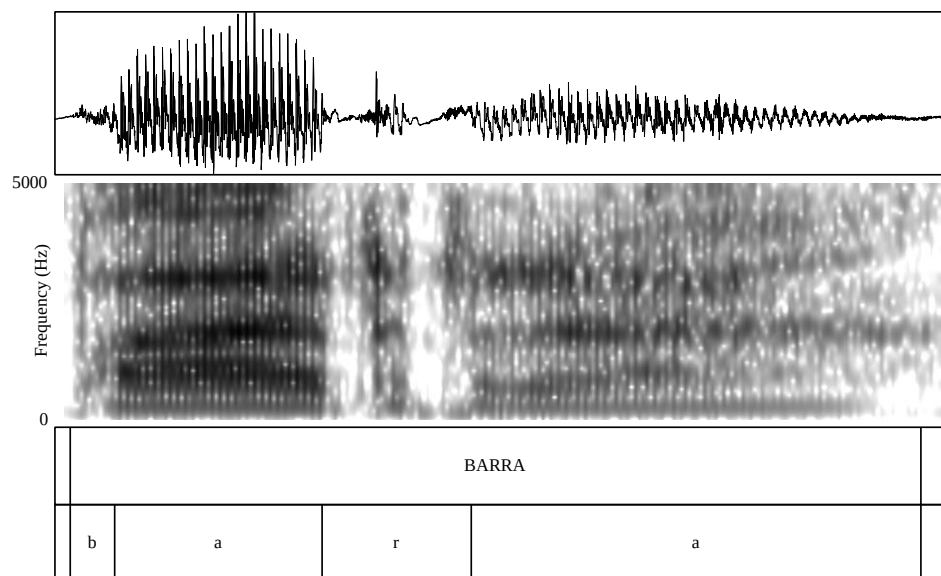


Figure 5. Intervocalic word-medial phonemic trill production containing one apical occlusion (plus r-coloring) in the word *barra* ('bar, counter,' trill duration = 71 ms)

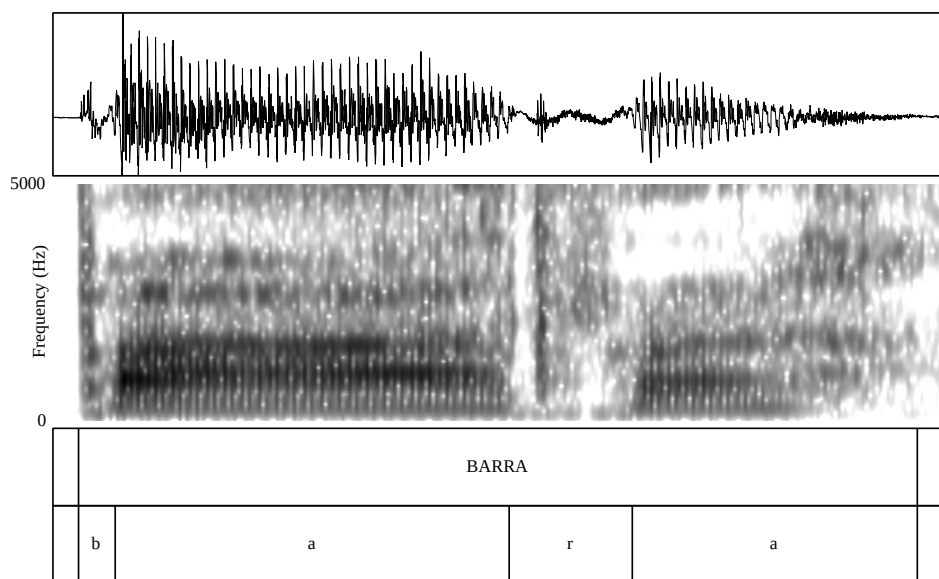
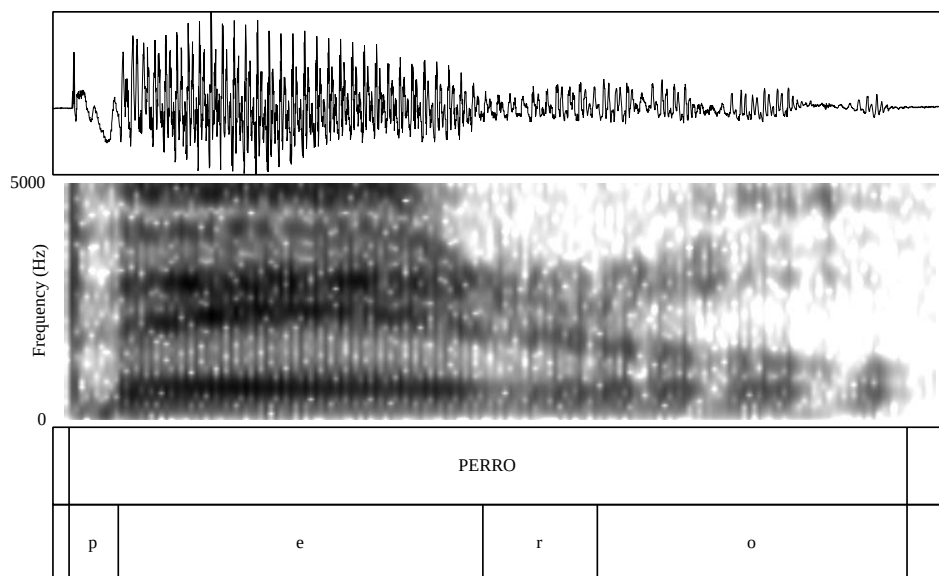


Figure 6. Intervocalic word-medial phonemic trill production containing zero apical occlusions in the word *perro* ('dog,' trill duration = 45 ms)



As in the analysis of phonemic trills, the overall duration of the word-medial intervocalic tap segment was extracted for each tap token. Inspection of the phonetic variants produced by the L2 learners and heritage speakers revealed that not all taps were produced with a clear apical occlusion. Following Henriksen (2015), the phonetic variants in this study were classified based on the degree of apical constriction as any of three types: true tap (TT), approximant tap (AT), and perceptual tap (PT).

Figure 7 provides an example of a TT that contains a clear break in the spectrogram along with reduced amplitude of the waveform. An AT was attested when there was an observable continuation of the formant structure from neighboring vowels and less reduction in the waveform as compared to the TT. In the articulation of the AT, the tip of the tongue approaches the alveolar ridge but without sufficient precision to create turbulent airflow. The formant structure in the AT is visible but not as clearly defined as that of neighboring vowels. An example of an AT is provided in Figure 8. Finally, a PT lacks distinguishable landmarks for segmenting the tap phoneme from surrounding vowels. The term ‘perceptual’ is used here following Henriksen (2015), since auditory analysis of these tokens suggests that a rhotic sound is produced between the adjacent vowels, even if it is not clearly visible. It is important to note that the calculation of tap duration in these examples is not possible since a rhotic segment is not visible in the acoustic signal. A PT is provided in Figure 9.

Figure 7. Intervocalic word-medial TT containing one apical occlusion in the word *moro* (‘moor,’ tap duration = 16 ms)

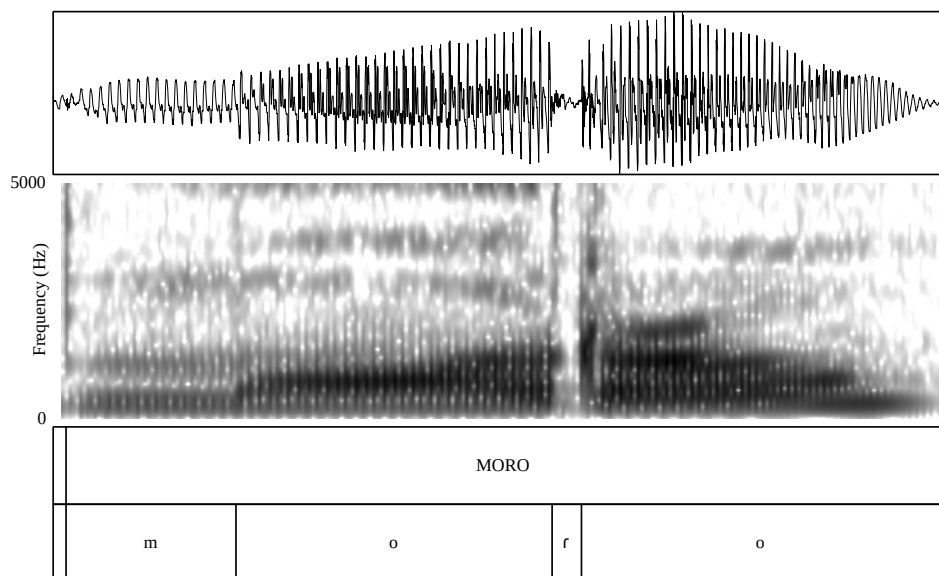
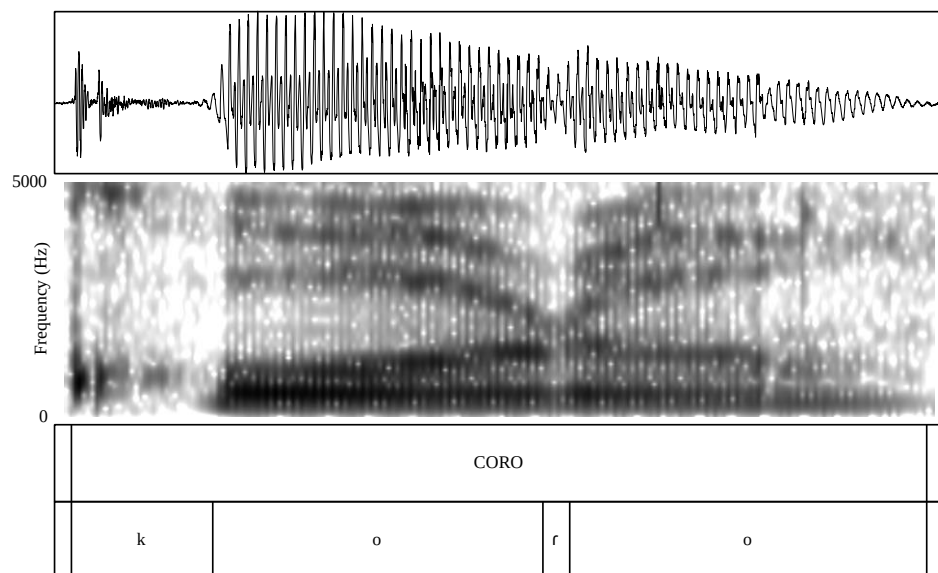
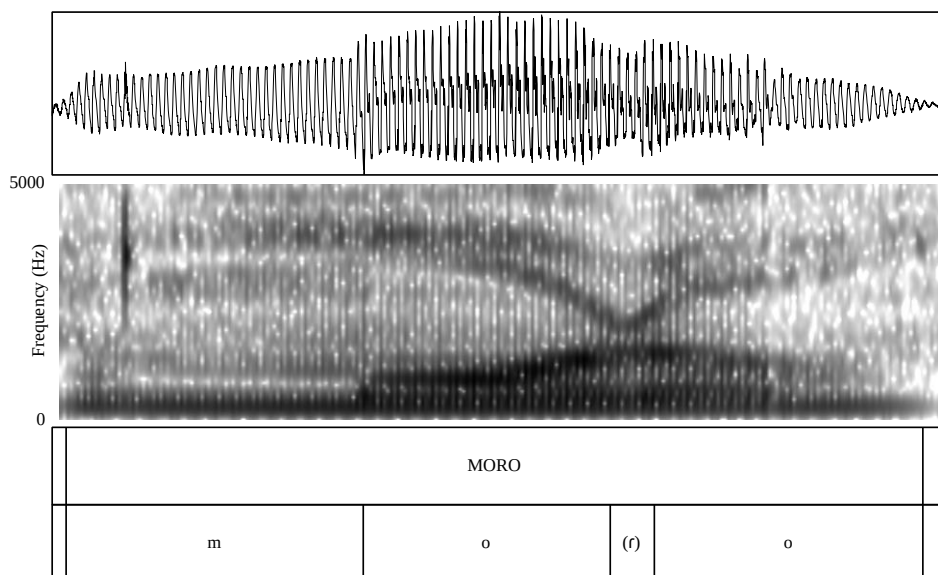


Figure 8. Intervocalic word-medial AT in the word *coro* ('choir,' tap duration = 12 ms)**Figure 9.** Intervocalic word-medial PT in the word *moro* ('moor')

In terms of the statistical analysis, datasets of by-subjects aggregates were created including the mean number of occlusions (i.e., trills) and mean duration (i.e., taps and trills) over subjects as a

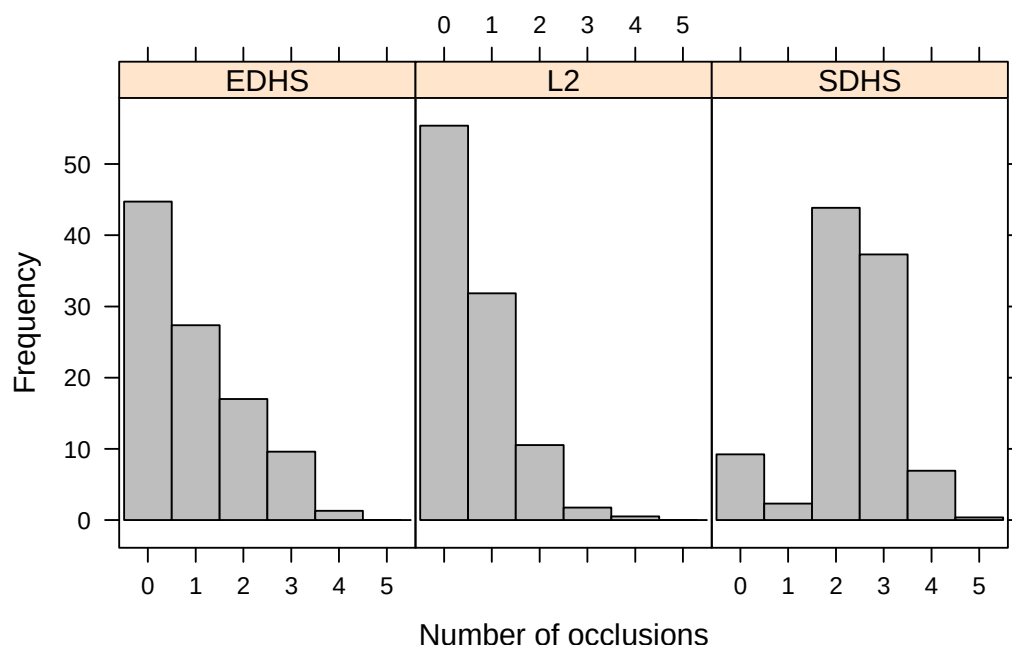
condition of word. Each dataset was submitted to a mixed-model ANOVA with participant as the random term (R Development Core Team, 2008). Models were developed for taps (i.e., duration) and trills (i.e., duration and number of occlusions) separately, which yielded outputs regarding the significance ($p=0.05$ threshold) of the fixed effect (i.e., speaker group). Group differences were further examined by means of Bonferroni-corrected, paired t-tests, and the alpha level was adjusted accordingly.

RESULTS

Phonemic Trill

The data for the phonemic trill were classified based on the number of occlusions and overall duration of the rhotic segment (in ms). The acoustic analysis revealed that SDHS produced more occlusions ($M = 2.31$, $SD = 0.99$) than EDHS ($M = 0.95$, $SD = 1.05$) and L2 learners ($M = 0.60$, $SD = 0.78$). A dataset of by-subject aggregates was created including the mean number of occlusions over subjects as a condition of word (20 values per participant). The occlusion data were analyzed through a mixed design ANOVA with *speaker group* (i.e., SPHS, EDHS, L2) as a between-subjects factor and *individual speaker* as a random effects factor. The model yielded a significant effect for *speaker group* ($F(2,57) = 40.29$, $p < 0.001$). Bonferroni-adjusted pairwise comparisons showed that there were significant differences in the number of occlusions in the trills produced by SDHS in comparison to EDHS (diff. = 1.36, $t(54) = 17.79$, $p < 0.001$), and also between SDHS and L2 learners (diff. = 1.71, $t(46) = 23.46$, $p < 0.001$), and between EDHS and L2 learners (diff. = .35, $t(93) = 5.84$, $p < 0.001$). Figure 10 provides the distribution of the number of occlusions (i.e., frequency of occurrence) for each speaker group.

Figure 10. Distribution of the number of occlusions for each speaker group

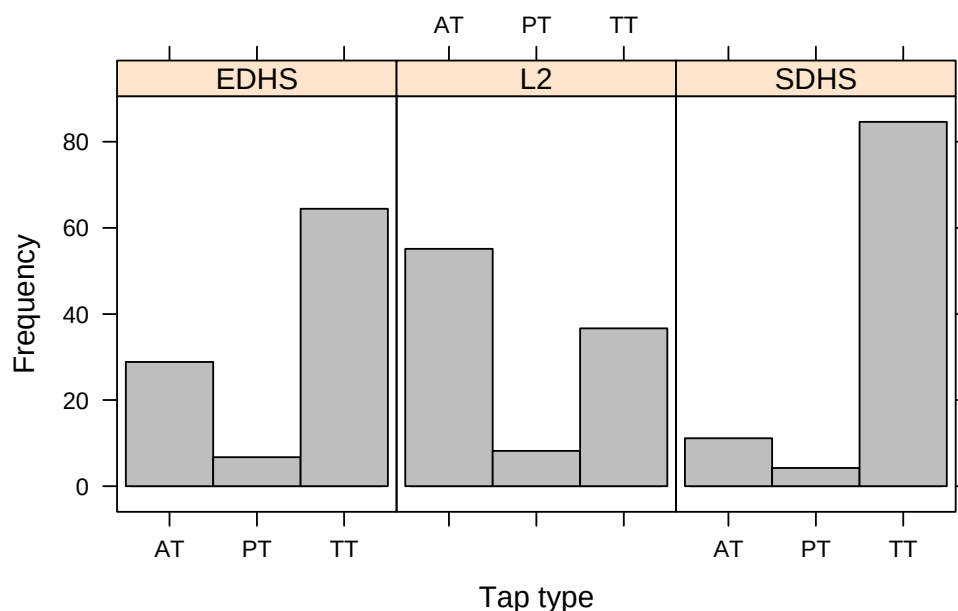


In terms of duration, SDHS produced phonemic trills with longer duration ($M = 90.25\text{ms}$, $SD = 23.24$) than EDHS ($M = 74.89\text{ms}$, $SD = 28.08$) and L2 learners ($M = 87.59\text{ms}$, $SD = 33.64$). A mixed design ANOVA with trill duration as the dependent variable also revealed a significant effect of *speaker group* ($F(2,57) = 4.25$, $p < 0.05$). Pairwise comparisons yielded significant differences in trill duration between SDHS and EDHS (diff. 15.36 , $t(60) = 8.16$, $p < 0.001$), and also between L2 learners and EDHS (diff. 12.7 , $t(76) = 6.13$, $p < 0.001$). However, there were no significant differences in duration between SDHS and L2 learners (diff. $= 2.66$, $t(65) = 1.18$, $p = 0.23$).

Phonemic Tap

The acoustic realization of the phonemic tap was categorized into three different types based on degree of segmental weakening: TT, AT, and PT, as in Henriksen (2015). The data indicated that the SDHS group favored the TT (84.6%, $N = 220$), followed by the AT (11.1%, $N = 29$), and then the PT (4.2%, $N = 11$). A similar order of preference was found in the productions of the EDHS, who produced the TT most frequently (64.4%, $N = 344$), followed by the AT (28.8%, $N = 154$), and the PT (6.7%, $N = 36$). Finally, a markedly different distribution was noted in the phonetic variants employed by the L2 learner group: AT (55.1%, $N = 215$), TT (36.6%, $N = 143$), and PT (8.2%, $N = 32$). Figure 11 provides the distribution of the phonetic variants of the phonemic tap for each speaker group. Recall that duration measurements were only extracted for ATs and TTs since PTs lack a clear acoustic correlate by which segmental duration can be measured. The duration measurements indicated that SDHS ($M = 45.12\text{ms}$, $SD = 27.65$) produced phonemic taps with a slightly longer duration than L2 learners ($M = 44.99\text{ms}$, $SD = 23.02$) and EDHS ($M = 37.58\text{ms}$, $SD = 16.04$). A mixed design ANOVA with tap duration as the dependent variable did not return a significant effect for *speaker group* ($F(2,57) = 1.47$, $p = 0.23$).

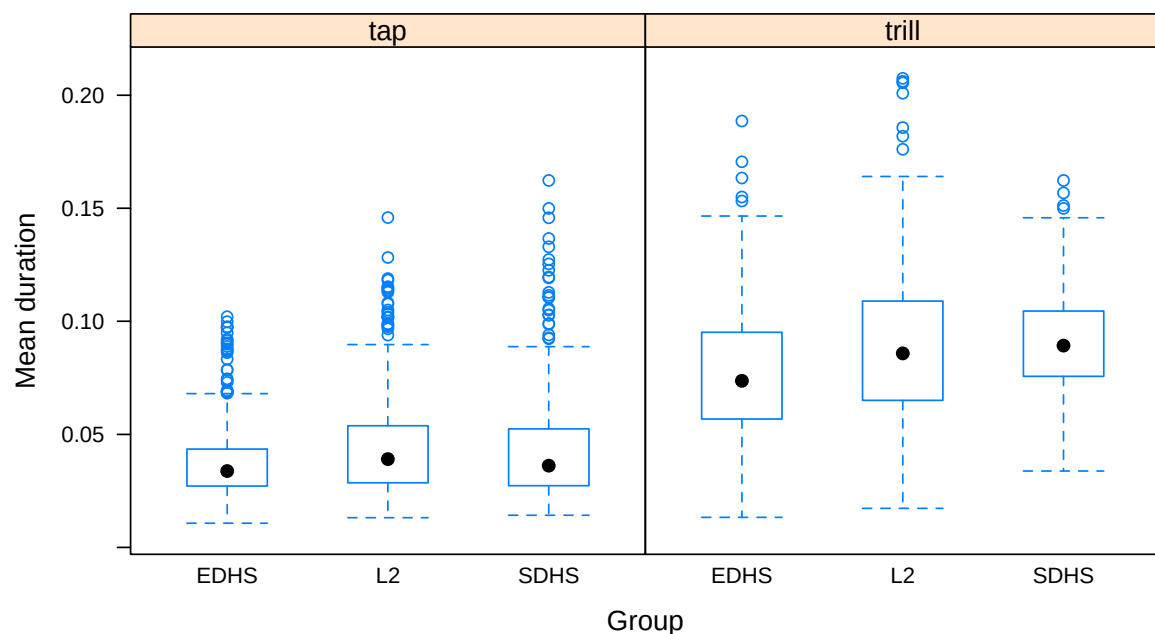
Figure 11. Distribution of the phonetic variants of the phonemic tap for each speaker group



Phonological Tap-trill Contrast

The acoustic analyses above revealed much variation in the articulation of the tap and trill phonemes by L2 learners, SDHS, and EDHS. In addition to comparing the rhotic production of groups of speakers that differ in age of acquisition, language dominance, and language use, one of the main goals of this production study is to examine the acoustic correlates of the tap-trill contrast in order to determine if these bilinguals are maintaining a robust rhotic contrast or if there is evidence of rhotic neutralization in the productions of these speakers (Hammond, 1999). The acoustic analyses revealed that many phonemic trill tokens, especially in the productions of L2 learners and EDHS, were not produced with the canonical two or more alveolar occlusions (Solé, 2002), but were produced as 1-occlusion or 0-occlusion trills instead. The acoustic realization of the trills by SDHS showed that they produced most of their tokens as a canonical trill with two or more occlusions. In order to determine if the rhotic contrast is maintained, at least in terms of duration, in the productions of the L2 learners and EDHS, a dataset of by-subject aggregates was created including the mean duration over subjects as a condition of rhotic type (2 values per participant). The duration data were analyzed through a mixed design ANOVA with *speaker group* (i.e., SPHS, EDHS, L2) as a between-subjects factor, *rhotic type* (i.e., tap, trill) as a within-subjects factor, and *individual speaker* as a random effects factor. The model yielded a significant effect for *speaker group* ($F(2,57) = 4.96, p < 0.05$) and *rhotic type* ($F(1,57) = 184.47, p < 0.001$), but no significant interaction between *speaker group* and *rhotic type* ($F(2,57) = 0.56, p = 0.66$). These results confirmed that all of the speaker groups realize the tap-trill contrast via segmental duration, and that only the SDHS maintained the contrast via segmental duration as well as number of lingual contacts. Figure 12 presents the duration values separately of the production of the phonemic tap and trill for each speaker group.

Figure 12. Duration values in the production of the phonemic tap and trill for each speaker group



DISCUSSION AND CONCLUSIONS

The present study examined the production of the Spanish phonological tap-trill contrast (/ɾ/-r/) by 60 Spanish-English bilinguals in Northern California. While there is a wealth of research that has examined dialectal variation of taps and trills (Bradley & Willis, 2012; Henriksen & Willis, 2010; Willis, 2007; Willis & Bradley, 2008) and the acquisition of the rhotic contrast by English-speaking L2 Spanish learners (Face, 2006; Major, 1986; Olsen, 2012; Reeder, 1998; Rose, 2010), there are only a few experimental studies that have investigated the rhotic contrast in the production of Spanish heritage speakers (Goodin-Mayeda, 2016; Henriksen, 2015; Kissling, 2016). One of the main contributions of the present study is that it examines which phonetic properties interact to achieve the phonological tap-trill contrast in the production of Spanish-English bilinguals who differ in proficiency, age of acquisition, language experience, and language dominance. Due to differences in the age of acquisition and language experience of the Spanish-English bilingual participants in this study, these speakers were classified as either L2 Spanish learners who had acquired Spanish later in life in a classroom setting and were dominant in English (their L1) or Spanish heritage speakers who had acquired both Spanish and English at an early age in a naturalistic setting. Based on their responses on the BLP (Birdsong, Gertken, & Amengual, 2012), the heritage speaker group was further divided into EDHS and SDHS.

This study posed several questions regarding the acoustic realization of the rhotic contrast by these Spanish-English bilinguals: do Spanish heritage speakers and L2 learners produce acoustically similar phonological trills? Do the acoustic properties of these trills match the canonical productions with two or more occlusions as reported in the literature? How do these bilinguals differ in their productions of phonemic taps? Finally, do these bilinguals maintain the phonological tap-trill contrast in Spanish or is this rhotic contrast being neutralized? In the case that the rhotic contrast is maintained, what are the acoustic correlates of this contrast in the production of L2 learners and heritage speakers?

The results from this production study indicated that there is much variation in the acoustic realization of the intervocalic taps and trills by the three speaker groups that were analyzed. Specifically, only the SDHS were found to produce mostly canonical phonological trills with two or three brief occlusions between the tongue apex and the alveolar ridge, whereas the EDHS and L2 learners produced the majority of trills with one or zero occlusions. A non-canonical production of the phonemic trill containing fewer than two apical closures has also been reported in the production of monolingual Spanish speakers (Bradley & Willis, 2012; Henriksen & Willis, 2010; Willis, 2007; Willis & Bradley, 2008), and first and second generation Spanish heritage speakers in Chicago (Henriksen, 2015).

The acoustic analysis of the phonemic tap showed that these groups also differed in the phonetic variants employed to pronounce the Spanish tap. In comparison to the production of the phonemic trills, the SDHS and EDHS groups showed a similar pattern, with a preference for the TT, followed by the AT, and then the PT. A different distribution emerged in the production of the L2 learner group, which displayed a more frequent production of the AT, followed by the TT, and the PT. Finally, a comparison of the tap and trill duration values indicated that each group produced significantly longer phonemic trills than taps. These results are in line with previous findings on monolingual and bilingual Spanish speakers showing that the contrast

between the tap and the trill is frequently realized via overall segmental duration, even if there is no consistent difference in the number of lingual contacts. In other words, rhotic duration seems to be the most robust correlate of the tap-trill contrast (Goodin-Mayeda, 2016; Henriksen, 2015).

Previous research has assumed that heritage speakers have a benefit in the acquisition of the heritage language sound system as a result of early exposure. However, several experimental studies have shown that there are measurable fine-grained differences at the phonetic level between the pronunciation of heritage speakers and that of monolingual speakers of the minority language (Bullock, 2009; Bullock & Gerfen, 2005). Furthermore, heritage language researchers have urged investigators to instrumentally test the assumption that heritage speakers maintain ‘good phonology’ in their minority language in order to shed light on the sound systems of heritage speakers in general (Polinsky & Kagan, 2007). The current study contributes to this effort by investigating the effects of language dominance on the production patterns of heritage speakers, an undoubtedly heterogeneous group. Even if these bilinguals maintain the rhotic contrast in Spanish, does their language experience affect their production patterns? Is language dominance a reliable factor in the examination of the acoustic realization of the rhotic contrast?

Language dominance refers to the “observed asymmetries of skill in, or use of, one language over the other” (Birdsong, 2014, p. 374), and this variable is especially relevant since language dominance has been shown to be an important predictor of the interaction patterns between the sound systems of both languages of the bilingual individual. For instance, recent studies have demonstrated that language dominance accounts for phonological transfer in speech production, perception, and processing (Amengual, 2016a, 2016b, 2016c; Amengual & Chamorro, 2015; Antoniou, Tyler, & Best, 2012; Bullock, Toribio, González, & Dalola, 2006; Simonet, 2014). Even though heritage speakers often have a high level of proficiency in their minority language, the perfectly balanced bilingual probably does not exist: heritage speakers, as any other bilinguals, have a dominant or stronger language (Cutler, Mehler, Norris, & Seguí, 1989; Flege, MacKay, & Piske, 2002). Regardless of being simultaneous or sequential bilinguals, heritage speakers typically become dominant in the majority language. Many heritage speakers, however, have a markedly different language experience with the minority language by maintaining a high frequency of use of this language and, as a result, present a markedly different profile from those bilinguals who become dominant in the majority language at the cost of the minority language. It is reasonable to assume that these bilinguals may be able to reduce the amount of cross-linguistic influence from the majority language while also preserving more robust language-specific phonemic contrasts in their heritage language.

Models of L2 phonological acquisition, such as the Speech Learning Model (SLM) (Flege, 1995), the Perceptual Assimilation Model (PAM) (Best, 1995), the Perceptual Assimilation Model for L2 Acquisition (PAM-L2) (Best & Tyler, 2007), and the Second Language Linguistic Perception model (L2LP) (Escudero, 2005, 2009) explain how L1 experience influences L2 sound learning and predict the relative difficulty L2 learners face when acquiring new or similar sounds. For instance, the SLM proposes that success in L2 production depends on the establishment of new phonetic categories of the L2 segments, and, critically, this success is based on the perceived similarity or dissimilarity between the L2 sounds and any existing L1 categories.

As stated in Flege (1995), “the more distant from the closest L1 sound an L2 sound is judged to be, the more likely it is that L2 learners –regardless of age – will establish a new category for the L2 sound” (p. 367). Considering findings from markedness theory and speech perception, the Spanish canonical trill should be acquired successfully as it is an acoustically salient sound to the L2 speaker and therefore the non-native speaker should be able to form a phonetic category that is distinguishable from other rhotic sounds in Spanish or English. However, the Spanish trill has also been shown to be difficult for L2 learners to produce, leading to large acoustic and articulatory variability. Unlike the trill, the tap is a sound that exists in English and should be more likely to be transferred to L2 Spanish, in spite of the status of the tap as an allophone of /t/ and /d/ in English and the difference in orthographic representation of tap in Spanish and English. Previous models do not account for the acoustic and articulatory variability of rhotic production because they do not consider the degree of difficulty involved in the acquisition of the multiple phonological and physiological characteristics of these L2 sounds. Consequently, as proposed in Colantoni and Steele (2008), models of L2 phonological acquisition must also incorporate phonetic constraints on production.

The SLM also postulates that early learners, as opposed to adult L2 learners, are capable of establishing additional phonetic categories for similar L2 sounds. Thus, the sounds of the L1 and L2 are less likely to interact in early bilinguals, such as the heritage speakers in the present study, than in late L2 learners. However, current L2 phonological acquisition models do not predict how language dominance will impact the acquisition of L2 sounds in the speech of bilinguals who have been exposed to the language at an early age, such as Spanish heritage speakers. In a recent study comparing the production of Spanish monolinguals, Spanish-speaking immigrants, and Spanish heritage speakers of a variety of Spanish from central Mexico, Kissling (2016) suggests that due to the heterogeneity of “heritage speakers,” the variability observed in acoustic data may make comparisons of this group to native speaker groups not meaningful, and that researchers might be better served to identify the underlying linguistic and social factors that explain their behavior, thereby circumscribing the “heritage” group more narrowly. The present study is a step forward in this direction: language dominance was operationalized as a multifaceted, gradient and dynamic construct by taking into account a variety of language-related variables, and the results showed that SDHS and EDHS produced acoustically distinct rhotic segments in Spanish. The results of this study corroborate that heritage speakers are a heterogeneous group in which variance is a result of language dominance; that is, EDHS and L2 learners are most likely to exhibit a modified system to maintain the intervocalic rhotic phonological contrast. These findings add to our understanding of the sources of variation in heritage and L2 pronunciation by investigating a largely understudied bilingual population that has traditionally been ignored in bilingual phonetic research. Additionally, the study provides further evidence that future work needs to consider proficiency, age of acquisition, language dominance, and other biographical non-linguistic variables that are particular to the experiences of these early bilinguals in order to better understand the phonological/phonetic systems of heritage speakers.

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NOTE

- [1.](#) In this context, *minority* and *majority* language refer to the language spoken by the minority or majority of the population of the U.S. as a whole. It is acknowledged that in the case of U.S. Latina/o communities, Spanish may be the majority language in terms of number of speakers (e.g., for instance, it is indeed spoken by the majority of the population in some counties in Texas and New Mexico), but nonetheless, it is the minority language in terms of economic, social or political prestige. For the purpose of this study, the minority language is the heritage language spoken at home (i.e., Spanish) and the majority language is English.